MODULE 06 · DOMAIN 1 - SNOWFLAKE AI DATA CLOUD FEATURES AND ARCHITECTURE

Cortex, Snowpark, Streamlit, Notebooks and ML

The largest block of genuinely new material on this exam, and the one older courses do not have.

DOMAIN 1 of 5 WEIGHT 31% MODULE 6 of 31

admin@sthithapragna.com

Every service here exists so the **model comes to the data**, rather than the data being exported to a model.

The Cortex landscape

- **AISQL functions.** LLM operations you call from SQL, row by row. Complete, classify, filter, aggregate, translate, sentiment.
- **Cortex Search.** Semantic and keyword retrieval over *unstructured* documents. The retrieval half of RAG.
- **Cortex Analyst.** Natural language to SQL over *structured* tables, grounded on a semantic model.
- **Document AI.** Structured field extraction from PDFs and images.

> ON THE EXAM
>
> Structured or unstructured is the first question to ask. It separates Analyst from Search immediately.

The AISQL functions

FUNCTION	WHAT IT DOES
AI_COMPLETE	Sends a prompt to a model and returns the completion. The general-purpose one.
AI_CLASSIFY	Sorts text into categories you supply. Multi-label, images and documents.
AI_FILTER	Returns true or false. Usable in SELECT, WHERE, and JOIN … ON.
AI_AGG	Aggregates a text column across rows. Not subject to context-window limits.
AI_PARSE_DOCUMENT	Extracts content from a document file.
AI_SENTIMENT, AI_TRANSLATE, AI_REDACT	Sentiment, translation, and PII scrubbing.

ON THE EXAM

AI_FILTER in a WHERE clause makes an LLM part of the query plan itself.

A naming change in progress

- The **AI_ prefixed functions are the current, canonical names**. Use them.
- The older **SNOWFLAKE.CORTEX namespace is documented as legacy**, to be deprecated by the end of 2026.
- Old to new: **COMPLETE becomes AI_COMPLETE**. CLASSIFY_TEXT becomes AI_CLASSIFY. PARSE_DOCUMENT becomes AI_PARSE_DOCUMENT.
- **Cortex Guard** filters unsafe model responses, and is built on Llama Guard.

> ON THE EXAM
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> The study guide says 'Cortex AI SQL' without naming functions. Expect either naming, and recognise both.

Cortex Search or Cortex Analyst?

CORTEX SEARCH

Unstructured retrieval

- Documents, contracts, tickets, articles
- Embeddings plus keyword ranking
- Returns relevant passages
- 'What does our returns policy say?'

CORTEX ANALYST

Structured question answering

- Tables, columns, metrics
- Natural language translated into SQL
- Requires a semantic model in YAML
- 'What were our top products last month?'

THE DOCUMENTED FAILURE MODE

Point a structured question at Search, or an unstructured one at Analyst, and you get an answer that is valid and useless.

What makes Cortex Analyst work

- It requires a **semantic model**, a lightweight YAML file. Similar in shape to a schema, but far richer.
- The semantic model carries **business definitions and metric handling** that a bare schema cannot express.
- The generated SQL **runs in your own virtual warehouse**, under **the same RBAC** as the underlying data.
- Snowflake states that Cortex Analyst **does not train on customer data**.

> ON THE EXAM
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> A thin semantic model produces SQL that runs cleanly and answers the wrong question. That is the practical risk.

Documents, agents, and fine-tuning

- **Document AI** and **AI_PARSE_DOCUMENT** extract structured fields from PDFs and images. Extraction, not retrieval.
- **Cortex Agents** orchestrate. They call Analyst and Search rather than replacing either.
- **Snowflake Intelligence** is the end-user surface sitting over the top of the agent layer.
- **Cortex Fine-tuning** adapts selected base models on your own data, without full training costs.

> ON THE EXAM
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> Document AI extracts fields into columns. Cortex Search finds passages. Both read documents, for different purposes.

Who is allowed to use Cortex

- Access needs **two** things: the **USE AI FUNCTIONS** account privilege, and a database role such as **SNOWFLAKE.CORTEX_USER**.
- **Both are granted to PUBLIC by default**, and PUBLIC is granted to every user. So by default, everyone can use Cortex.
- To restrict it: **revoke from PUBLIC as ACCOUNTADMIN**, then grant to the specific roles that should have it.
- **Database roles cannot be granted directly to a user.** Grant to a role, then assign that role.
- Narrower roles exist: **CORTEX_ANALYST_USER**, CORTEX_AGENT_USER, CORTEX_EMBED_USER, AI_FUNCTIONS_USER.

> ON THE EXAM
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> 'Granted to PUBLIC by default' is the governance answer. Nobody enabled Cortex; it was on from the start.

Cross-region inference

- Not every model is available in every region. If yours is not, the call fails unless cross-region inference is enabled.
- Controlled by **CORTEX_ENABLED_CROSS_REGION**. Default is **DISABLED**. Values include ANY_REGION and specific scopes such as AWS_US.
- **Account level only**, not user or session. Only **ACCOUNTADMIN** can set it, and **ORGADMIN cannot**.
- **Credits are consumed in the requesting region**, and there are **no data egress charges**.
- Your stored data never moves. Only the prompt and response travel, transiently, encrypted in transit.

> ON THE EXAM
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> A compliance-flavoured question about data residency and AI usually turns on this parameter.

How the AI services are billed

SERVICE	CHARGED AS
AISQL functions	AI Credits per million tokens processed. The rate depends on the model chosen.
Cortex Analyst	Its generated SQL runs in your virtual warehouse, so ordinary warehouse credits.
Cortex Agents	AI Credits per million tokens, and additive across every service the agent calls.
ML functions	Compute for training and prediction, plus storage for the trained model instances.
Notebooks	Warehouse credits, or compute pool credits on the Container Runtime.

ON THE EXAM

Model choice is a cost decision. A larger model on the same prompt can cost several times more per million tokens.

That is the AI. The rest of the module is about running your own code inside Snowflake.

Snowpark

- Libraries and runtimes for **Python, Java and Scala**, running inside Snowflake.
- A **DataFrame API**. Operations are **lazy** and get translated into SQL that runs on a warehouse.
- That translation is the point. Work is **pushed down** to the engine rather than pulling rows to a client.
- Used to write **UDFs, UDTFs and stored procedures** in those languages. Covered again in module 22.
- Available in **Standard Edition**, like everything else in this module.

> ON THE EXAM
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> Snowpark is not a separate engine. A DataFrame chain becomes SQL, and you can see it in Query History.

A memory rule that surprises people

- **Standard warehouses have a lower Snowpark memory limit, and that limit does NOT increase with warehouse size.**
- For more memory per node you need a **Snowpark-optimized warehouse**. That is what they exist for.
- **Snowpark-optimized memory does not increase above Medium** either. Going bigger buys cores, not headroom.
- Guidance: train on a **dedicated warehouse** with no concurrent work. Prototype on an **X-Small standard** to control cost.

> ON THE EXAM
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> An out-of-memory failure on a standard warehouse is not fixed by resizing it. That is a scenario waiting to happen.

Streamlit and Container Services

- **Streamlit in Snowflake** builds and hosts data applications from Python, next to the data, with no separate web server.
- Apps inherit Snowflake **authentication and RBAC**. Sharing an app is a grant, not a deployment.
- **Snowpark Container Services** runs containerised workloads on Snowflake-managed **compute pools**, including GPUs.
- That is what you reach for when the workload needs a language or library Snowflake does not host natively.

> ON THE EXAM
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> Streamlit is also the migration path for retired Snowsight Dashboards, via the conversion tool from module 5.

Notebooks: two runtimes

WAREHOUSE RUNTIME

Runs on a virtual warehouse

- Fastest to start
- The familiar warehouse Python environment
- CPU only
- SQL analytics and general work

CONTAINER RUNTIME

Runs on compute pools

- Powered by Snowpark Container Services
- Install packages from PyPI or internal repos
- CPU or GPU, with ML packages pre-installed
- Machine learning and deep learning

THE DETAIL PEOPLE MISS

For BOTH runtimes, SQL and Snowpark queries always execute on the warehouse. Only the notebook's own Python differs.

The four ML functions

FUNCTION	WHAT IT ANSWERS
Forecasting	What will this metric be in future, based on past trends? Time series.
Anomaly Detection	Which values differ from what we would expect? Time series.
Classification	Which class does each row belong to? Does NOT require a timestamp.
Top Insights	Which dimensions and values are driving the change in this metric? Root cause.

ON THE EXAM

Top Insights was formerly called Contribution Explorer. Older material and some questions still use the old name.

The rest of Snowflake ML

- **Model Registry.** The central repository for machine learning models across the organisation.
- **Feature Store.** A repository of data transformations that can be reused to train models.
- Container Runtime gives **automatic lineage capture** and Model Registry integration.
- All of it is **Standard Edition**. No part of the AI or ML surface is edition-gated.

> ON THE EXAM
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> Registry stores models. Feature Store stores the transformations that feed them. Two repositories, two contents.

Which tool for which job

THE REQUIREMENT	THE ANSWER
Summarise or classify text held in a table column	AISQL functions
Let staff ask questions of a document library	Cortex Search
Let staff ask questions of sales tables in plain English	Cortex Analyst plus a semantic model
Pull named fields out of scanned invoices	Document AI or AI_PARSE_DOCUMENT
Process large data in Python without moving it	Snowpark
Train on GPUs, or install arbitrary packages	Notebooks on Container Runtime, or Container Services
Predict a metric, or explain why it moved	ML functions: Forecasting, or Top Insights

ON THE EXAM

Seven rows covering the module. The second and third are the pair worth over-learning.

> A legal team holds eight years of supplier contracts as PDFs in a Snowflake stage, with a directory table over them. Lawyers want to ask questions such as which contracts contain a change-of-control clause, and to be shown the relevant passages.
>
> Separately, the finance team wants to ask questions such as which suppliers we spent the most with last quarter, against ordinary tables.

Which services fit these two requirements?

A. Cortex Analyst for both, since it answers natural-language questions.

B. Cortex Search for the contracts, and Cortex Analyst for the finance tables.

C. Document AI for the contracts, and Cortex Search for the finance tables.

D. AI_COMPLETE for both, prompted with the relevant rows each time.

◆ PAUSE AND DECIDE

Which services fit these two requirements?

A. Cortex Analyst for both.

Analyst generates SQL over structured tables. It has nothing to run against a library of PDFs.

B. Search for contracts, Analyst for finance.

Unstructured retrieval with passages returned, and structured question answering. Each requirement gets the matching service.

C. Document AI for contracts, Search for finance.

Both halves inverted. Document AI extracts known fields; Search does not query structured tables.

D. AI_COMPLETE for both.

Would need the right content in the prompt already. Finding that content is the actual problem, and it is what Search does.

THE SIGNAL

'Shown the relevant passages' is the tell for Search. Retrieval returns passages; extraction returns fields.

> A data science team runs a Snowpark Python job that trains a model over a wide feature table. On a Large standard warehouse it fails with an out-of-memory error.
>
> They resize the warehouse to 2X-Large and rerun. It fails identically. They resize to 4X-Large. It fails identically again, and the credits consumed have risen sharply.

Why is resizing not helping, and what should they do?

A Keep scaling. Memory grows with warehouse size, so 5X-Large or 6X-Large will eventually succeed.

B The Query Acceleration Service should be enabled to offload the memory-intensive portion.

C Enable multi-cluster scaling, which distributes the job's memory across clusters.

D Standard warehouse memory limits do not increase with size. Use a Snowpark-optimized warehouse.

◆ PAUSE AND DECIDE

Why is resizing not helping, and what should they do?

A. Keep scaling to 5X-Large or 6X-Large.

What they have already tried twice. Standard warehouse memory limits do not increase with size, so this never succeeds.

B. Query Acceleration Service.

QAS offloads scan work for eligible SQL queries. It does not raise the memory ceiling for Snowpark code.

C. Multi-cluster scaling.

Multi-cluster adds clusters for concurrency. A single job still runs within one cluster's memory.

D. Standard memory does not scale. Use Snowpark-optimized.

Documented directly. Snowpark-optimized warehouses provide more memory per node, which is why the type exists.

THE SIGNAL

Snowpark-optimized memory does not rise above Medium either. Bigger buys cores, not headroom, on both warehouse types.

A team wants to develop a deep-learning model in a Snowflake Notebook. They need a specific open-source library that is not in Snowflake's package list, and the training step requires GPU acceleration.

They also run several large SQL aggregations in the same notebook to build the training set, and are worried these will be slow in a notebook environment.

Which runtime should they choose, and what happens to the SQL?

A Container Runtime. The SQL and Snowpark queries still execute on the warehouse regardless of runtime.

B Warehouse Runtime, because SQL must run on a warehouse and the runtime determines that.

C Container Runtime, but the SQL aggregations will now run on the compute pool and be slower.

D Warehouse Runtime with a Snowpark-optimized warehouse, which provides GPU access.

◆ PAUSE AND DECIDE

Which runtime should they choose, and what happens to the SQL?

A Container Runtime. SQL still runs on the warehouse.

Gives PyPI installs and GPU compute pools, and for both runtimes SQL and Snowpark always execute on the warehouse.

B Warehouse Runtime.

Cannot install arbitrary packages and offers no GPU. Both stated requirements fail.

C Container Runtime, but SQL moves to the pool.

Right runtime, wrong consequence. This is exactly the misconception the documentation corrects.

D Warehouse Runtime with Snowpark-optimized.

Snowpark-optimized warehouses provide more memory, not GPUs. GPUs come from Container Services compute pools.

THE SIGNAL

C is the best distractor: correct choice, invented downside. The runtime governs the notebook's Python, not its SQL.

> A subscription business has seen monthly conversion rate fall sharply. The data has dimensions for channel, region, device, plan tier and campaign. Leadership wants to know which combinations of those dimensions account for most of the fall.
>
> A analyst suggests building a forecasting model to project where conversion will go next.

Which ML function actually answers the question asked?

A Forecasting, as suggested, since it models the metric over time.

B Anomaly Detection, to flag the months where conversion was abnormal.

C Top Insights, which finds the dimensions and values driving the change.

D Classification, to sort customers into converting and non-converting classes.

◆ PAUSE AND DECIDE

Which ML function actually answers the question asked?

A Forecasting.

Projects where the metric is going. Leadership asked why it already moved, which is a different question.

B Anomaly Detection.

Identifies which months were abnormal. They already know the fall happened; they want the cause.

C Top Insights.

Purpose-built for root cause: which dimensions and values are affecting the metric in surprising ways.

D Classification.

Predicts a class per row. It does not attribute a change in an aggregate metric to segments.

NAMING

Top Insights was formerly Contribution Explorer. A question may use either name for the same function.

What to carry into the next module

- **Search is unstructured, Analyst is structured.** Analyst needs a semantic model in YAML.
- **AI_ functions are current**; SNOWFLAKE.CORTEX is legacy. Both granted to PUBLIC by default.
- **Standard warehouse memory does not grow with size.** Out-of-memory means Snowpark-optimized.
- **Container Runtime for GPUs and custom packages.** SQL runs on the warehouse either way.
- **ML functions:** Forecasting, Anomaly Detection, Classification, Top Insights.

> ON THE EXAM
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> Next: the Domain 1 drill. Eight scenarios across everything from modules 1 to 6, and no new content.